

ENV 4120

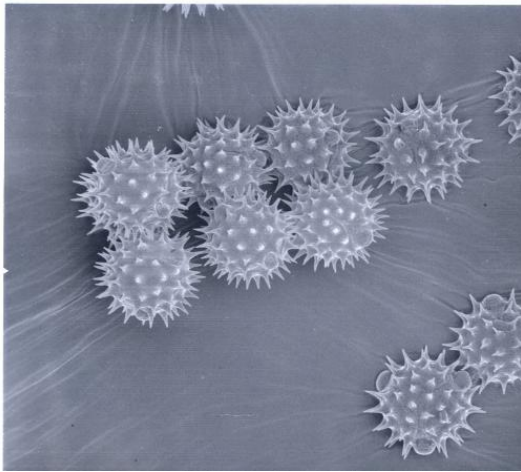
Air Pollution and Hazardous Waste Control

Today's Topics

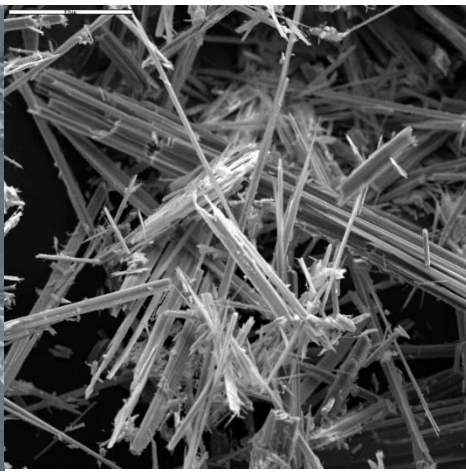
- Sources and interactions of atmospheric PM
- Expectation
 - Understand size distributions and mode (3-mode model) of PM, major sources of PMs and their interactions

Particulate matter

- PM are everywhere in the environment
- Vary greatly in source, size and composition



pollen



asbestos



volcanic ash

Terminology

- Particulate matter (PM): complex mixture of very small particles and liquid droplets
- Aerosol: ensemble of suspended PMs
- Total suspended particles (TSP) - all suspended PMs
- Coarse particles: PMs with diameter $> 2.5 \mu\text{m}$
- Inhalable coarse particles: PM_{10} - $\text{PM}_{2.5}$
- Fine particles: $\text{PM}_{2.5}$
- Ultrafine particles: $\text{PM}_{0.1}$

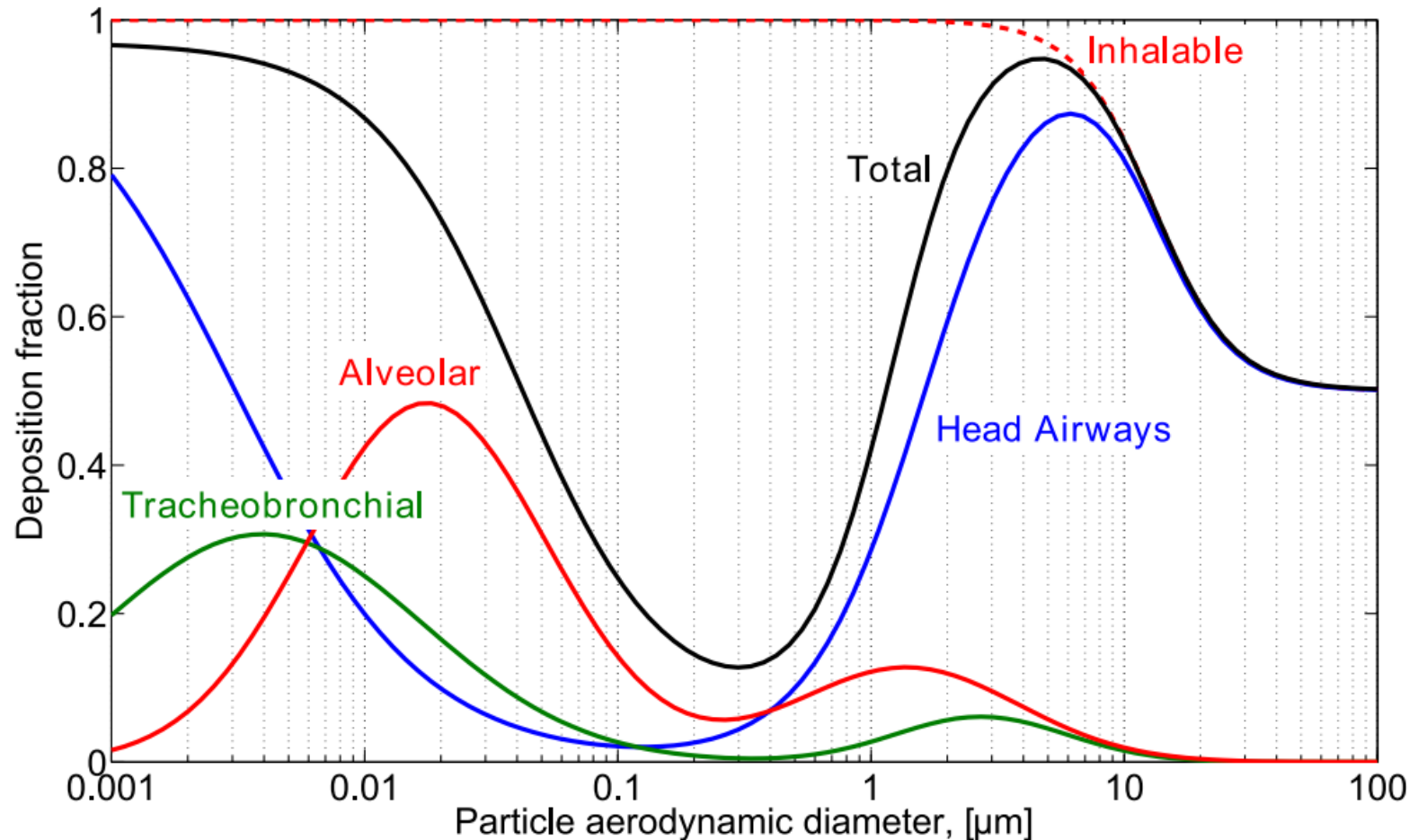
Concentration Metrics

- Three metrics depending on purpose
 - Number concentration
 - e.g. light scattering and absorption
 - Area concentration
 - e.g. surface reactions
 - Area of a spherical particle: $a_p = 4\pi r^2 = \pi D_p^2$
 - Mass concentration
 - e.g. health effect due to mass dose
 - Volume of a spherical particle: $v_p = 4/3\pi r^3 = 1/6\pi D_p^3$

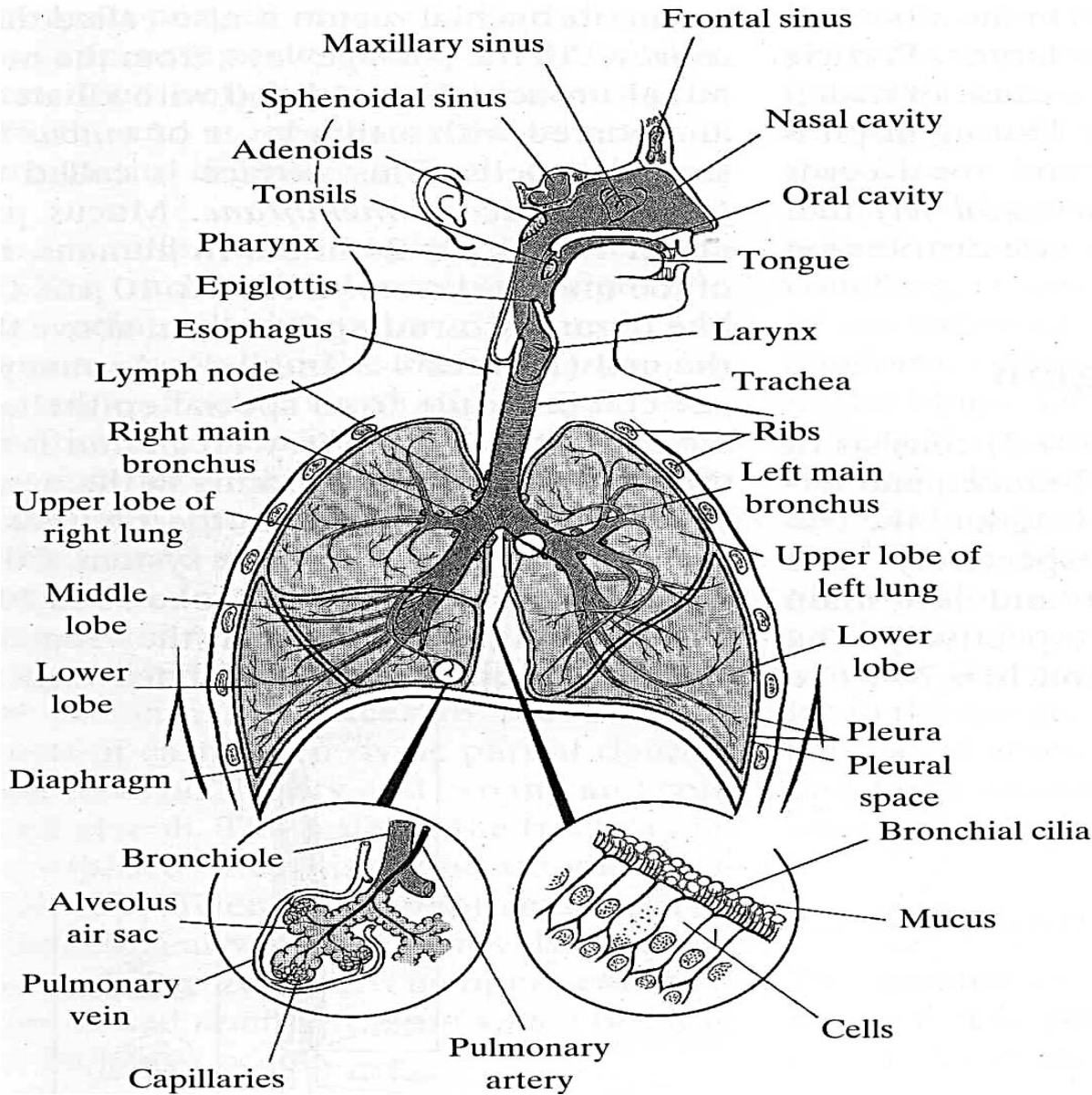
Particulate matter

- For aerosol, size matters
 - Different size particle behavior differently
 - Smaller size particles generally:
 - Suspend in the air longer
 - Goes deeper inside lungs (sort of)
- Size for non-spherical aerosol
 - Most used: aerodynamic diameter (D_a)
 - The diameter of a round water droplet (1g/cm^3) that has the same settling velocity as the particle

Particle deposition in human respiratory tract



The respiratory system



Sources and Control of Air Pollution, Heinsohn and Kabel, Prentice-Hall, 1999

The respiratory system

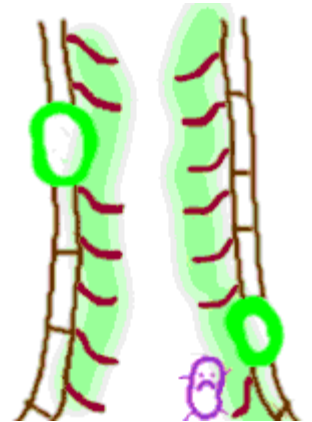
- Consist of upper airway, tracheobronchial tree and pulmonary region
- Overall function
 - Gas to blood exchange
 - Supply the body with O_2 for metabolism
 - Remove waste CO_2
 - Removal of other volatile toxicants
 - Physical immune response
 - Eliminating of viruses, etc before they get to the blood

The respiratory system

- Upper airway
 - Nose to vocal chords
 - Warm and humidifies air
 - Filtering of particles by hairs
 - Aerodynamic impaction of particles (winding passages)
 - Adsorption in mucus (which is swallowed or spit out)
 - Sneeze reflex
 - Olfaction (smell)

The respiratory system

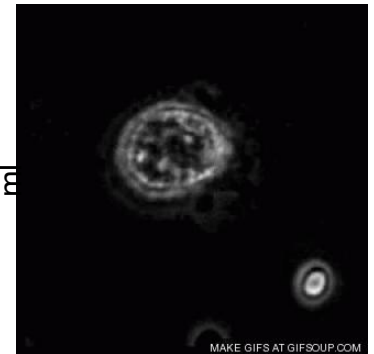
- Tracheobronchial tree
 - Air transport
 - Bronchoconstriction – capture foreign substances
 - Mucociliary escalator – beating cilia move mucus and foreign substances toward mouth/nose
 - Cough reflex



The respiratory system

- Pulmonary region
 - Respiratory bronchioles, alveolar ducts, alveoli (air sacs)
 - Site of air to blood gas exchange (mass transfer by molecular diffusion)
 - 50-100 m² of gas exchange surface
 - Phagocytosis - ingestion of deposited matter by phagocytes and macrophages (from white blood cells)
 - Absorption to lymphatic system

<http://gifsoup.com/view/2305747/phagocytosis.html>



MAKE GIFS AT GIFSOUP.COM

PM in the respiratory system

- Particles must be deposited to have an effect
- Major factor for particulate behavior is their aerodynamic properties
- Depend on size (and orientation), shape, density
 - Not all particles can be inhaled
 - Some inhaled particles are exhaled

PM in the respiratory system

- Upper airways:
 - Capture larger particles ($D > 5\text{-}10\ \mu\text{m}$); clear in $< 1\ \text{d}$
 - Capture very small particles ($D < 0.01\ \mu\text{m}$)
 - Mouth breathing allows larger PM to penetrate deeper in lungs
- Pulmonary region
 - Deposition of small particles ($0.01 < D < 4\ \mu\text{m}$)
 - Clearance takes weeks to months
 - Particles $< 0.1\ \mu\text{m}$ are readily absorbed by tissues

Mode of PM

- Essentially size bins
- Here we use three modes:
 - Nucleation mode ($D < 0.1\mu\text{m}$)
 - Accumulation mode ($2.5\mu\text{m} > D > 0.1\mu\text{m}$)
 - Coarse mode ($D > 2.5\mu\text{m}$)
- Bin # and size ranges vary depending on purpose

Mode of PM

- Nucleation mode ($D < 0.1\mu\text{m}$)
 - Mainly combustion PMs, and particles formed in the atmosphere by gas-to-particle conversion.
 - Quickly grow into accumulation mode.
 - Largest number concentration
- Accumulation mode ($2.5\mu\text{m} > D > 0.1\mu\text{m}$)
 - Still small but growth can be slow
 - Large surface area concentration

Mode of PM

- Coarse mode ($D > 2.5\mu\text{m}$)
 - Mainly primary emissions
 - Natural emissions mainly coarse mode
 - Also comes from mechanical process
 - Agriculture, surface mining etc
 - Generally sink down within a few days
 - Largest mass concentration
- $D > 10\mu\text{m}$ generally not a health concern

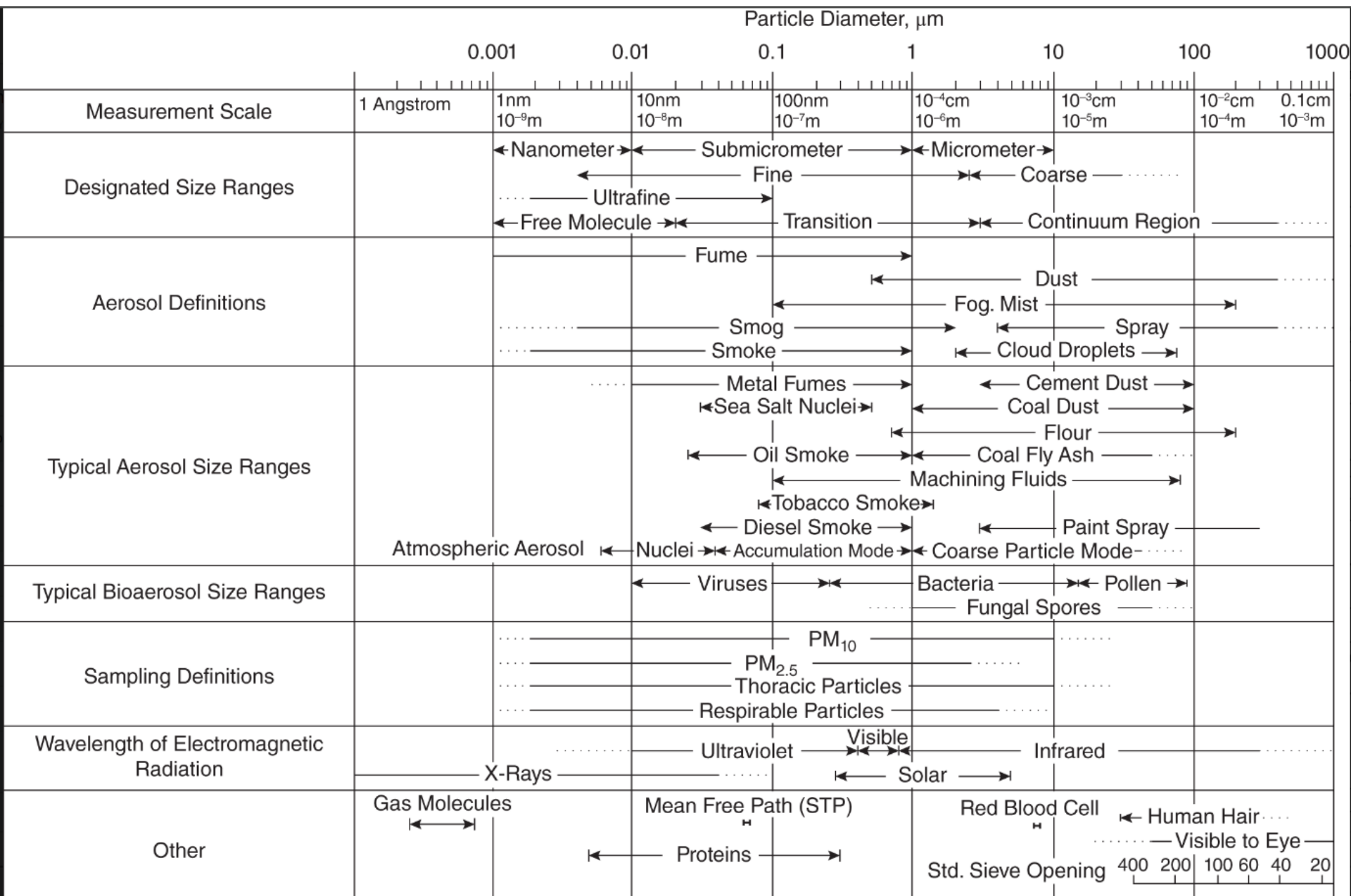
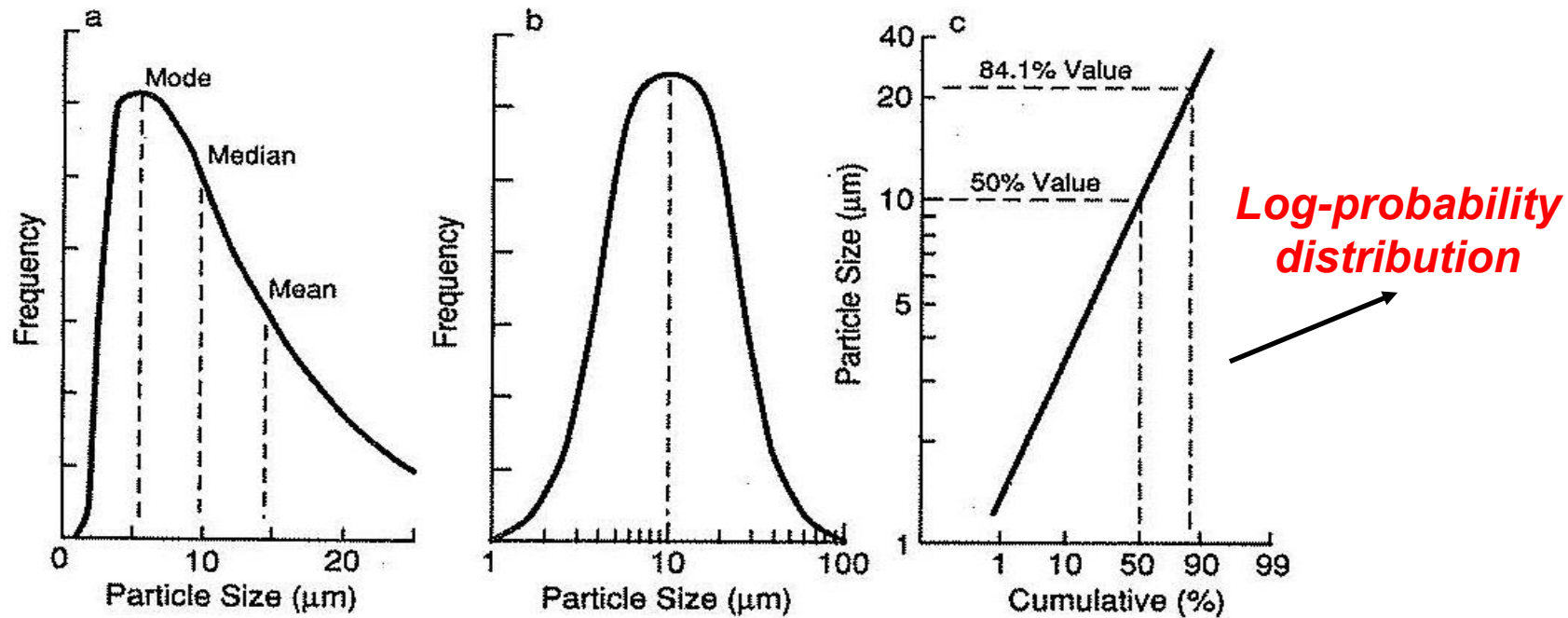


FIGURE 1.6. Hinds & Zhang. 2023. Particle size ranges and definitions for aerosols.

Size distribution of aerosols

- A sample of aerosols usually contains particles of different sizes

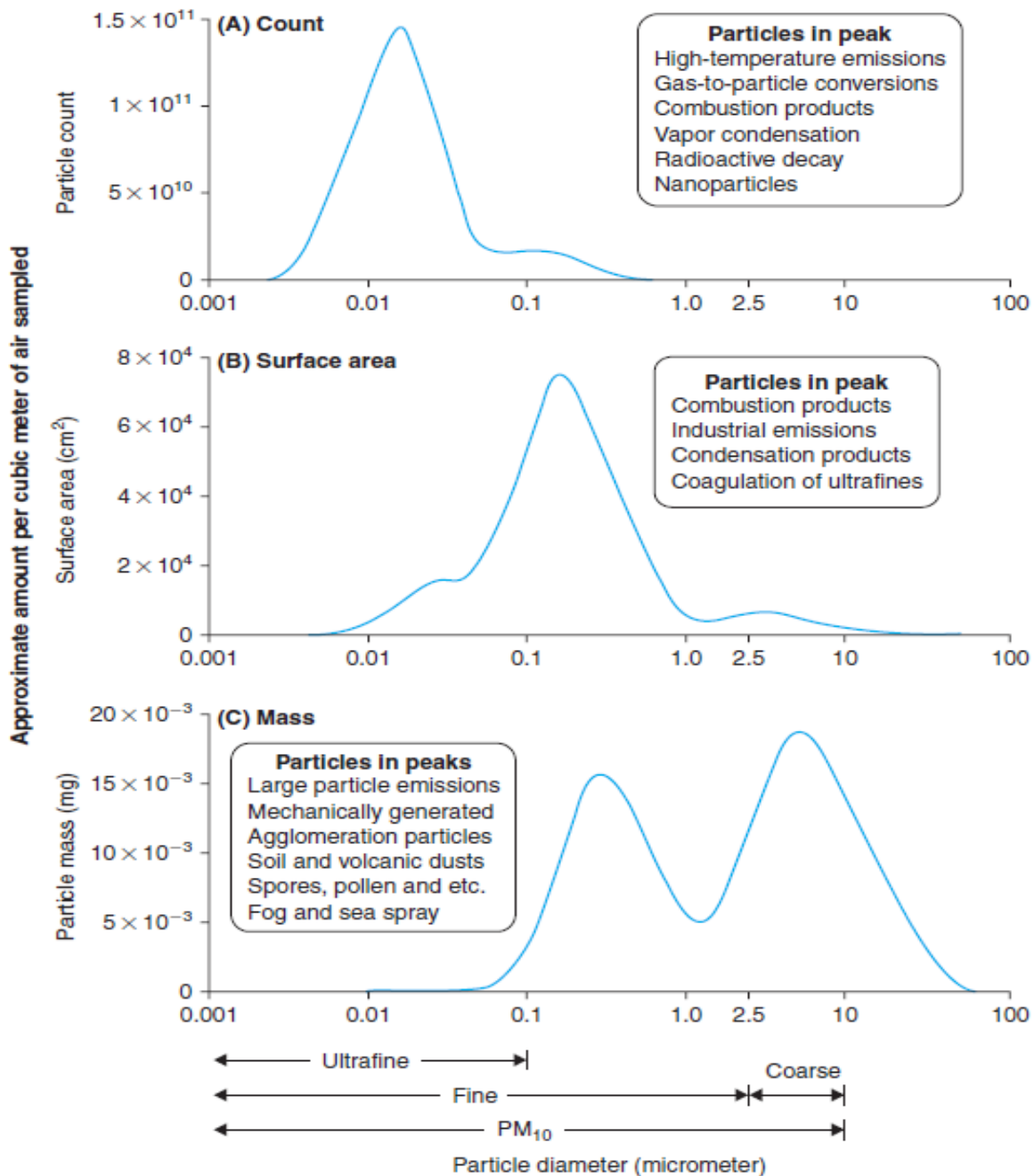


Differential and cumulative size distributions.

Source: Lippmann, M. (2000) 'Introduction and background'. In: M. Lippmann (ed.) *Environmental Toxicants*, Wiley, New York.

(from Tiwary)

Example count, surface area and mass distributions of different size aerosols



(Phalen, 2013)

Source & compositions

Table 14.1 Sources and Estimates of Global Emissions of Atmospheric Aerosols [a\)](#), [b\)](#).

	Natural Source		Anthropogenic Source		Total	
Particle Type	Range	Average	Range	Average	Range	Average
Biogenic primary organic	15–70	50	—	—	15–70	50
Biogenic secondary organic	3–83	11	—	—	3–83	11
Biogenic VOCs	835–1000	900	—	—	835–1000	900
Biomass burning organic	26–70	44	12–270	90	38–340	134
Black carbon (soot)	—	—	8–14	12	8–14	12
Carbonaceous	—	—	15–90	29	15–90	29
Industrial dust	—	—	40–130	100	40–130	100
Mineral dust	1000–2150	1731	50–250	150	1050–2400	1881
Nitrates	12–27	25	90–118	95	102–145	120
Sea salt	3000–20,000	7068	—	—	3000 – 20,000	7068
Stratospheric volcanic	2–17	10	—	—	2–17	10
Sulfates	107–374	165	50–122	73	157–496	238
Tropospheric volcanic	4–90	30	—	—	4–90	30
Total Emissions	5004–23,881	10034	265–994	408	5269–24,875	10442

[a\)](#) Adapted from Tomasi et al. (2016).

[b\)](#) Units are in Tg/yr [10^6 metric tons/yr].

Sea spray

- Largest natural source aerosol source
- Mainly coarse
- Wind → bubbles → spray
- Main composition
 - Seawater with some sodium chloride (NaCl)



Soil dust

- Natural soil dust
 - Main compositions: mineral and organic materials
 - Mainly coarse
 - Desert and damaged land cover area
- Anthropogenic soil dust
 - Agriculture
 - Re-suspended dust
 - Construction, demolition etc.



Volcanic dust

- Mainly coarse silicate materials
- Secondary PMs can form from volcanic gases

Mount Saint Helen, 1980



Biomass burning

- Biomass burning PMs containing plant debris, ash, soil dust, soot etc., and are mainly coarse
- Also emit gases such as phenol, cause secondary aerosol formation
- Study has shown that 40% of global black carbon emissions come from biomass burning



Fossil fuel combustion

- Burning of solid, liquid, or gas fossil fuels
 - Mainly soot, sulfate, fly ash, metals
- Many fine PMs, some coarse (fly ash)
- Diesel vehicle emit 2-20 times than gas vehicle
 - Without control 10-100 times
- NASA visualization of global aerosol/PM:
https://www.youtube.com/watch?v=oRsY_UviBPE
- A snap shot image from NASA:
<https://www.nasa.gov/image-feature/just-another-day-on-aerosol-earth>

Interaction of particles

- Coagulation
- Impaction/interception
- Conversion between size modes

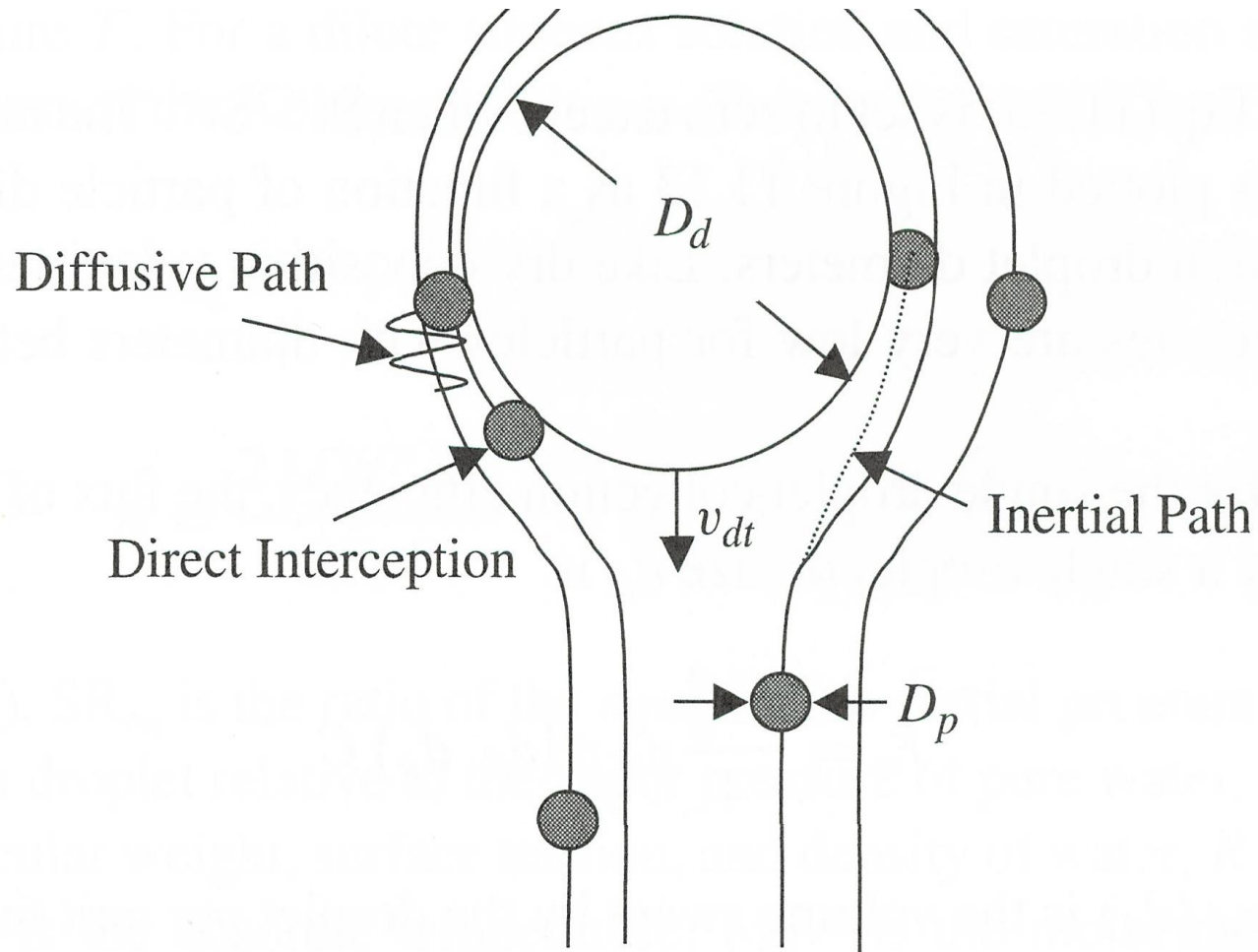
Coagulation

- Two particles collide and stick together
- Number concentration ↓ & particle size ↑
 - Same mass concentration
- When Brownian diffusion dominates particle motion
 - Thermal coagulation
- When external forces dominates particle motion
 - Kinematic coagulation

Impaction/interception

- Scavenging of particles by other particles or surfaces
- Impaction
 - Particles have inertia and can't follow the gas streamlines
 - Direct impact on other particles or surface
- Interception
 - Particle remains on streamline but is large enough to contact anyway

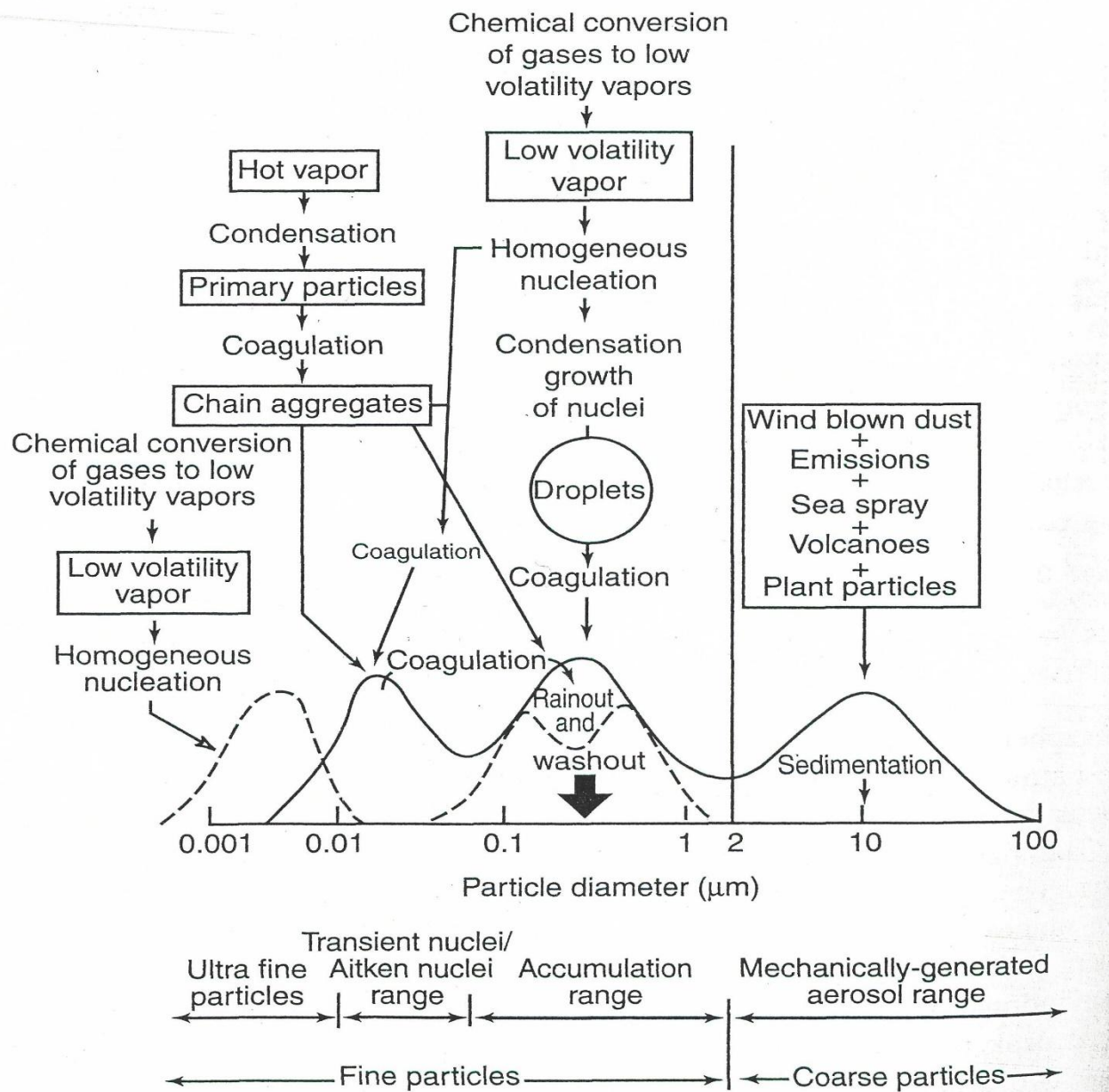
Impaction/interception



Conversion between size modes

- The size distribution of aerosol will change over time and space
 - Due to all of the processes mentioned previously
- Particle itself: condensation/evaporation
- Many processes affect particle size distribution

Processes affecting particle size distribution



Source and processed of atmospheric aerosols at different size modes (Godish 2003).
 Solid line: three mode model by Whitby & co-worker; Dash line: added ultrafine mode with sometime observed accumulation model distribution. Vertical axis: maybe number, area or mass. This figure is for illustration only